$$D = P^{-1}AP$$

$$= \frac{1}{-3}\begin{bmatrix} 2 & -1 \\ -1 & -1 \end{bmatrix}\begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}\begin{bmatrix} -1 & 1 \\ 1 & 2 \end{bmatrix}$$

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$A^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$= \frac{1}{-3}\begin{bmatrix} 2(1)-1(4) & 2(2)-1(3) \\ -1(1)-1(4) & -1(2)-1(3) \end{bmatrix}\begin{bmatrix} -1 & 1 \\ 1 & 2 \end{bmatrix}$$

$$= \frac{1}{-3}\begin{bmatrix} -2 & 1 \\ -5 & -5 \end{bmatrix}\begin{bmatrix} -1 & 1 \\ 1 & 2 \end{bmatrix}$$

$$= -\frac{1}{3}\begin{bmatrix} -2(-1)+1(1) & -2(1)+1(2) \\ -5(-1)+-5(1) & -5(1)-5(2) \end{bmatrix}$$

$$= -\frac{1}{3}\begin{bmatrix} 3 & 0 \\ 0 & -15 \end{bmatrix}$$

$$D = \begin{bmatrix} -1 & 0 \\ 0 & 5 \end{bmatrix}$$

$$A^n = PD^nP^{-1}$$

$$A^4 = \begin{bmatrix} -1 & 1 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 5^4 \end{bmatrix} -\frac{1}{3}\begin{bmatrix} 2 & -1 \\ -1 & -1 \end{bmatrix}$$

CO - 2 | 05/02/25

Differential Calculus

① $\frac{d}{dx}(c) = 0$

② $\frac{d}{dx}(x^n) = n \cdot x^{n-1} \longrightarrow \frac{d}{dx}\left(\frac{1}{x}\right) = -\frac{1}{x^2}$

③ $\frac{d}{dx}(k \cdot f(x)) = k \cdot f'(x)$

④ $\frac{d}{dx}(e^{f(x)}) = e^{f(x)} \cdot f'(x)$

⑤ $\frac{d}{dx}(\log f(x)) = \frac{1}{f(x)} \cdot f'(x)$

⑥ $\frac{d}{dx}(k_1 f(x) \pm k_2 g(x)) = k_1 f'(x) + k_2 g'(x)$

⑦ $\frac{d}{dx}(uv) = uv' + u'v$

⑧ $\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{vu' - uv'}{v^2}$

⑨ $\frac{d}{dx}(\sin x) = \cos x$; $\frac{d}{dx}(\cos x) = -\sin x$

⑩ $\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$; $\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$

Partial Derivatives

When the given function $u = f(x,y)$ is a multiple valued function, then we can apply partial derivative in place of ordinary derivative, by for finding rate of change of 'u'.

*If $u = f(x, y)$ then (First - Order)

(i) The partial derivative of u w.r.t 'x' is defined as ordinary derivative of u w.r.t 'x' by treating 'y' as constant. It is denoted by $\frac{\partial u}{\partial x}$ (or) $\frac{\partial f}{\partial x}$ (or) u_x (or) $f_x = p$

(ii) The partial derivative of 'u' w.r.t 'y' is defined as ordinary derivative of 'u' w.r.t 'y' by treating 'x' as constant. It is denoted by $\frac{\partial u}{\partial y}$ (or) $\frac{\partial f}{\partial y}$ (or) u_y (or) $f_y = q$

Notation:- (Second - Order)

(1) $\frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x}\left[\frac{\partial u}{\partial x}\right] = u_{xx} = f_{xx} = r$

(2) $\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial}{\partial x}\left[\frac{\partial u}{\partial y}\right] = u_{xy} = f_{xy} = s$

(or)

$\frac{\partial u}{\partial y \partial x} = \frac{\partial}{\partial y}\left[\frac{\partial u}{\partial x}\right] = u_{yx} = f_{yx} = s$

$\therefore \boxed{u_{xy} = u_{yx}}$

(3) $\frac{\partial^2 u}{\partial y^2} = \frac{\partial}{\partial y}\left[\frac{\partial u}{\partial y}\right] = u_{yy} = f_{yy} = t.$

Session-9

① Determine First & second order partial derivatives of the function $f(x,y) = x^2y + \sin x + \cos y$ and also $f_{xy} = f_{yx}$

Sol:- $u = f(x,y) = x^2y + \sin x + \cos y$

$\rightarrow \frac{\partial u}{\partial x} = \frac{\partial}{\partial x}[x^2y + \sin x + \cos y]$

$= \frac{\partial}{\partial x}(x^2y) + \frac{\partial}{\partial x}(\sin x) + \frac{\partial}{\partial x}(\cos y)$

$\frac{\partial u}{\partial x} = y(2x) + \cos x + 0$

$\therefore \frac{\partial u}{\partial x} = 2xy + \cos x = u_x = p.$

$\rightarrow \frac{\partial u}{\partial y} = \frac{\partial}{\partial y}[x^2y + \sin x + \cos y]$

$= \frac{\partial}{\partial y}(x^2y) + \frac{\partial}{\partial y}(\sin x) + \frac{\partial}{\partial y}(\cos y)$

$= x^2 \cdot 1 + 0 - \sin y$

$\therefore \frac{\partial u}{\partial y} = x^2 - \sin y = u_y = q$

$\rightarrow \frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x}\left[\frac{\partial u}{\partial x}\right] = \frac{\partial}{\partial x}[2xy + \cos x]$

$= \frac{\partial}{\partial x}(2xy) + \frac{\partial}{\partial x}(\cos x)$

$= 2y - \sin x$

$\therefore \frac{\partial^2 u}{\partial x^2} = 2y - \sin x = u_{xx} = r$

$$\rightarrow \frac{\partial^2 u}{\partial x \partial y} = \frac{\partial}{\partial x}\left[\frac{\partial u}{\partial y}\right]$$

$$= \frac{\partial}{\partial x}\left[x^2 - \sin y\right]$$

$$= \frac{\partial}{\partial x}(x^2) - \frac{\partial}{\partial x}(-\sin y)$$

$$= 2x - 0$$

$$= 2x$$

$$\therefore \frac{\partial^2 u}{\partial x \partial y} = 2x = u_{xy} = s$$

$$\rightarrow \frac{\partial^2 u}{\partial y^2} = \frac{\partial}{\partial y}\left[\frac{\partial u}{\partial y}\right]$$

$$= \frac{\partial}{\partial y}\left[x^2 - \sin y\right]$$

$$= \frac{\partial}{\partial y}(x^2) - \frac{\partial}{\partial y}(-\sin y)$$

$$= 0 - \cos y$$

$$\therefore \frac{\partial^2 u}{\partial y^2} = -\cos y = u_{yy} = t.$$

$$u_{yx} = \frac{\partial^2 u}{\partial y \partial x} = \frac{\partial}{\partial y}\left[\frac{\partial u}{\partial x}\right]$$

$$= \frac{\partial}{\partial y}\left[2xy + \cos x\right]$$

$$= 2x(1) + 0$$

$$= 2x$$

$$\boxed{\therefore u_{xy} = u_{yx}}$$

② Determine First & Second order partial derivatives of $u = x^3 + y^4 + 4ax^2y$ at the point (1, 1).

Sol:- $\rightarrow \frac{\partial u}{\partial x} = \frac{\partial}{\partial x}\left[x^3 + y^4 + 4ax^2y\right]$

$= 3x^2 + 0 + 8ayx$

$u_x = \frac{\partial u}{\partial x} = 3x^2 + 8ayx.$

$u_{x(1,1)} = 3(1)^2 + 8a(1)(1)$

$= 3 + 8a$

$\rightarrow \frac{\partial u}{\partial y} = \frac{\partial}{\partial y}\left[x^3 + y^4 + 4ax^2y\right]$

$= 0 + 4y^3 + 4a$

$= 4y^3 + 4ax^2$

$\frac{\partial u}{\partial y} = 4(ax^2 + y^3)$

$u_{y(1,1)} = 4(a(1)^2 + (1)^3)$

$= 4a + 4.$

$\rightarrow \frac{\partial^2 u}{\partial x \partial y} = \frac{\partial}{\partial x}\left[\frac{\partial u}{\partial y}\right]$

$= \frac{\partial}{\partial x}\left[4y^3 + 4ax^2\right]$

$= \frac{\partial}{\partial x}(4y^3) + \frac{\partial}{\partial x}(4ax^2)$

$+$

$0 + 8ax$

$= 8ax = \frac{\partial^2 u}{\partial x \partial y_{(1,1)}}$

$= 8a$

$$\rightarrow \frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x}\left[\frac{\partial u}{\partial x}\right]$$

$$= \frac{\partial}{\partial x}\left[3x^2 + 8ayx\right]$$

$$\frac{\partial^2 u}{\partial x^2} = 6x + 8ay$$

$$\frac{\partial^2 u}{\partial x^2}_{(1,1)} = 6(1) + 8a(1)$$

$$= 6 + 8a \,//$$

$$\rightarrow \frac{\partial^2 u}{\partial y^2} = \frac{\partial}{\partial y}\left[\frac{\partial u}{\partial y}\right]$$

$$= \frac{\partial}{\partial y}\left[4y^3 + 4ax^2\right]$$

$$= 12y^2 + 0$$

$$\frac{\partial^2 u}{\partial y^2}_{(1,1)} = 12y^2$$

$$= 12(1)^2$$

$$\frac{\partial^2 u}{\partial y^2} = 12 \,//$$

(3) verify if $u_{xy} = u_{yx}$ for $u = \sin^{-1}\left(\frac{x}{y}\right)$

$$\frac{\partial u}{\partial x} = \frac{\partial}{\partial x}\left[\sin^{-1}\left(\frac{x}{y}\right)\right] \quad \sin^{-1}(x)$$

$$= \frac{1}{\sqrt{1-\left(\frac{x}{y}\right)^2}} \cdot \frac{\partial}{\partial x}\left(\frac{x}{y}\right)$$

$$= \frac{1}{y}(1) \cdot \frac{1}{\sqrt{1-\frac{x^2}{y^2}}} \cdot \cancel{= \frac{-x}{y^2} \cdot \frac{y}{\sqrt{y^2-x^2}}} = \cancel{\frac{x}{y\sqrt{y^2-x^2}}}$$

$$\frac{1}{\cancel{y}} \cdot \frac{\cancel{y}}{\sqrt{y^2-x^2}} = \frac{1}{\sqrt{y^2-x^2}}$$

$$u_y = \frac{\partial u}{\partial y} = \frac{\partial}{\partial y}\left[\sin^{-1}\left(\frac{x}{y}\right)\right] = \frac{1}{\sqrt{1-\left(\frac{x}{y}\right)^2}} \frac{\partial}{\partial y}\left(\frac{x}{y}\right)$$

$$= -\frac{x}{y^2} \cdot \frac{y}{\sqrt{y^2-x^2}}$$

$$= \frac{-x}{y\sqrt{y^2-x^2}}$$

$$u_{xy} = \frac{\partial^2 u}{\partial x \partial y} = \frac{\partial}{\partial x}\left[\frac{\partial u}{\partial y}\right]$$

$$= \frac{\partial}{\partial x}\left[\frac{-x}{y\sqrt{y^2-x^2}}\right]$$

$$= -\frac{1}{y} \frac{\partial}{\partial x}\left[\frac{x}{\sqrt{y^2-x^2}}\right]$$

$$= -\frac{1}{y}\left[\frac{\sqrt{y^2-x^2}\,(1) - (x)\,\frac{1}{2\sqrt{y^2-x^2}} \cdot \frac{\partial}{\partial x}(y^2-x^2)}{(y^2-x^2)}\right]$$

$$-\frac{1}{y}\left[\frac{\sqrt{y^2-x^2} + \frac{2x^2}{2\sqrt{y^2+x^2}}}{(y^2-x^2)^2}\right] = -\frac{1}{y}\left[\frac{\frac{2(y^2-x^2)+2x^2}{2\sqrt{y^2+x^2}}}{(y^2-x^2)}\right]$$

$$-\frac{1}{y}\left[\frac{\frac{\cancel{2}(y^2-x^2)+\cancel{2}x^2}{2\sqrt{y^2+x^2}}}{(y^2-x^2)}\right] = -\frac{1}{\cancel{y}}\left[\frac{2\,\cancel{y^2}}{(y^2-x^2)^{3/2}}\right]$$

$$= \frac{-y}{(y^2-x^2)^{3/2}}$$

$$u_{yx} = \frac{\partial^2 u}{\partial y \partial x} = \frac{\partial}{\partial y}\left[\frac{\partial u}{\partial x}\right]$$

$$= \frac{\partial}{\partial y}\left[\frac{1}{\sqrt{y^2-x^2}}\right]$$

$$= \frac{\partial}{\partial y}(y^2-x^2)^{-1/2}$$

$$= -\frac{1}{2}(y^2-x^2)^{-3/2}\frac{\partial}{\partial y}(y^2-x^2)$$

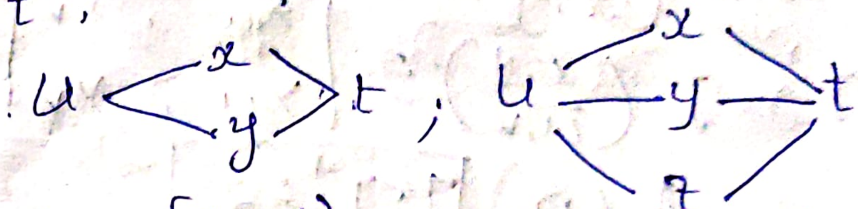
$$= -\frac{1}{2}(y^2-x^2)^{-3/2}(2y-0)$$

$$= \frac{-y}{(y^2-x^2)^{3/2}}$$

$$\boxed{u_{xy} = u_{yx}}$$

Total derivative and Jacobian:- 06/02/25

Total Derivative:- If 'u' is a function in terms of 'x' and 'y' and x & y are functions interms of 't' then $\frac{du}{dt}$ is called total derivative of u w.r.t 't'.



Note:- $u = f(x,y)$ where x & y are diff functions where $x = \phi(t), y = \phi(t)$.

Chain Rule:-

(1) $$\boxed{\frac{du}{dt} = \frac{\partial u}{\partial x}\cdot\frac{dx}{dt} + \frac{\partial u}{\partial y}\cdot\frac{dy}{dt}}$$ → 2 variable.

(2) $$\boxed{\frac{du}{dt} = \frac{\partial u}{\partial x}\cdot\frac{dx}{dt} + \frac{\partial u}{\partial y}\cdot\frac{dy}{dt} + \frac{\partial u}{\partial z}\cdot\frac{dz}{dt}}$$

3 variable

Jacobian:-

If u and v are two functions in terms of x & y, then the jacobian of u,v w.r.t x & y is defined as

$$J\left(\frac{u,v}{x,y}\right) = \begin{vmatrix} u_x & u_y \\ v_x & v_y \end{vmatrix}$$

$\begin{matrix} u \\ v \end{matrix} > x, y$ $\quad \frac{\partial(u,v)}{\partial(x,y)} = J$

Note:- $\begin{matrix} u \\ v \\ w \end{matrix} > x, y, z$

$$J\left(\frac{u,v,w}{x,y,z}\right) = \begin{vmatrix} u_x & u_y & u_z \\ v_x & v_y & v_z \\ w_x & w_y & w_z \end{vmatrix}$$

$$\frac{\partial(u,v,w)}{\partial(x,y,z)} = J$$

Note:-

If $J = J\left(\frac{u,v}{x,y}\right)$ and

$$J' = J\left(\frac{x,y}{u,v}\right) = \begin{vmatrix} x_u , & x_v \\ y_u , & y_v \end{vmatrix}$$

then $\boxed{J \cdot J' = 1}$

Session - 1:-

① Find the value of $\frac{du}{dt}$ where $u = y^2 - 4ax$ with $x = at^2$, $y = 2at$.

Sol:-

$$\begin{tikzcd} & x \arrow[dr] & \\ u \arrow[ur] \arrow[dr] & & t \\ & y \arrow[ur] & \end{tikzcd}$$

$$\frac{du}{dt} = \frac{\partial u}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial u}{\partial y} \cdot \frac{dy}{dt} \rightarrow ①$$

$$\frac{\partial u}{\partial x} = u_x = \frac{\partial}{\partial x}(y^2 - 4ax)$$

$$= \frac{\partial}{\partial x}(y^2) - \frac{\partial}{\partial x}(4ax)$$

$$= 0 - 4a(1)$$

$$\frac{\partial u}{\partial x} = 4a = -4a$$

$$u_y = \frac{\partial u}{\partial y} = \frac{\partial}{\partial y}\left[y^2 - 4ax\right]$$

$$= 2y - 0 \quad = 2y$$

$$\frac{dx}{dt} = \frac{d}{dt}\left[at^2\right] = a(2t)$$

$$\frac{dx}{dt} = 2at$$

$$\frac{dy}{dt} = \frac{d}{dt}\left[2at\right] = 2a(1)$$

$$\frac{dy}{dt} = 2a$$

$$\rightarrow \frac{\partial u}{\partial x} = (-4a)(2at) + 2y(2a)$$

$$= -8a^2t + 4ay$$

$$= -8a^2t + 4a(2at)$$

$$= \cancel{-8a^2t} + \cancel{8a^2t}$$

$$= 0$$

(2) If $u = \cos\left(\frac{x}{y}\right)$ with $x = e^t, y = t^2$ $\frac{du}{dt} = ?$

$$\frac{\partial u}{\partial x} = u_x = \frac{\partial}{\partial x}\left(\cos\left(\frac{x}{y}\right)\right)$$

$$\frac{\partial u}{\partial x} = u_x = -\sin\left(\frac{x}{y}\right)\frac{\partial}{\partial x}\left(\frac{x}{y}\right) = -\frac{1}{y}\sin\left(\frac{x}{y}\right)$$

$$\frac{\partial u}{\partial y} = u_y = \frac{\partial}{\partial y}\left(\cos\left(\frac{x}{y}\right)\right)$$

$$= -\sin\left(\frac{x}{y}\right)\frac{\partial}{\partial y}\left(\frac{x}{y}\right)$$

$$= -\sin\left(\frac{x}{y}\right)x\left(-\frac{1}{y^2}\right)$$

$$= \frac{x}{y^2}\sin\left(\frac{x}{y}\right)$$

$$\frac{dx}{dt} = \frac{d}{dt}[e^t] \qquad \frac{dy}{dt} = \frac{d}{dt}[t^2]$$

$$\frac{dx}{dt} = e^t \qquad \frac{dy}{dt} = 2t$$

$$\frac{\partial u}{\partial t} = \frac{\partial u}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial u}{\partial y} \cdot \frac{dy}{dt}$$

$$= \frac{-1}{y} \sin\left(\frac{x}{y}\right) \cdot e^t + \frac{x}{y^2} \sin\left(\frac{x}{y}\right) \times 2t$$

$$= \frac{1}{y} \sin\left(\frac{x}{y}\right)\left[-e^t + \frac{2tx}{y^2}\right]$$

$$= \frac{1}{t^2} \sin\left(\frac{e^t}{t^2}\right)\left[-e^t + \frac{2\cancel{t} \cdot e^t}{t^{\cancel{2}}}\right]$$

$$\frac{\partial u}{\partial t} = \frac{1}{t^2} \sin\left(\frac{e^t}{t^2}\right)\left[e^t\left(-1 + \frac{2}{t}\right)\right]$$

(3) Evaluate $J\left(\frac{u,v}{x,y}\right)$ where $u = x^2 - 2y$
$v = 3x + y$

Sol:- $J = J\left(\frac{u,v}{x,y}\right) \begin{vmatrix} u_x & u_y \\ v_x & v_y \end{vmatrix}$

$u = x^2 - 2y \Rightarrow u_x = \frac{\partial}{\partial x}(x^2 - 2y)$

$$u_x = 2x$$

$$u_y = \frac{\partial}{\partial y}(x^2 - 2y)$$

$$u_y = -2$$

$$V_x = \frac{\partial}{\partial x}(3x+y) = 3$$

$$V_y = \frac{\partial}{\partial y}(3x+y) = 1$$

$$J = \begin{vmatrix} 2x & -2 \\ 3 & 1 \end{vmatrix}$$

$$= 2x(1) - (-2)(3)$$

$$= 2x + 6$$

Session - 10

12/02/25

⑥ Evaluate $\frac{\partial(u,v,w)}{\partial(x,y,z)}$ at $(1,0,0)$

from $u = x^2 - 2y$

$v = x + 2y + z^2$

$w = x - 2y^2 + 3z$

Sol:-

$u_x = x^2 - 2y$

$= \frac{\partial}{\partial x}(x^2 - 2y)$

$\boxed{u_x = 2x}$

$u_y = x^2 - 2y$

$\boxed{u_y = -2}$

$u_z = \frac{\partial}{\partial x}(x^2 - 2y)$

$\boxed{u_z = 0}$

$V_x = \frac{\partial}{\partial x}(x + 2y + z^2)$

$\boxed{V_x = 1}$

$V_y = \frac{\partial}{\partial y}(x + 2y + z^2)$

$\boxed{V_y = 2}$

$V_z = \frac{\partial}{\partial x}(x + 2y + z^2)$

$\boxed{V_z = 2z}$

$w_x = \frac{\partial}{\partial x}(x - 2y^2 + 3z)$ | $w_y = (x-2y^2+3z)$ | $w_z = \frac{\partial}{\partial z}(x-2y^2+3z)$

$\boxed{w_x = 1}$ | $w_y = -4y$ | $\boxed{w_z = 3}$

$$\begin{vmatrix} u_x & u_y & u_z \\ v_x & v_y & v_z \\ w_x & w_y & w_z \end{vmatrix} = \begin{vmatrix} 2x & -2 & 0 \\ 1 & 2 & 2z \\ 1 & -4y & 3 \end{vmatrix}$$

$= 2x[6+8yz] + 2[3-2z]$

$= 12x + 16xyz + 6 - 4z = 12 + 6 = 18$

(5) $x = r\cos\theta$, $y = r\sin\theta$

$$\frac{\partial(x,y)}{\partial(r,\theta)} = \begin{vmatrix} x_r & x_\theta \\ y_r & y_\theta \end{vmatrix}$$

$x_r = \frac{\partial}{\partial r}(r\cos\theta)$ | $x_\theta = \frac{\partial}{\partial \theta}[r\cos\theta]$

$x_r = \cos\theta$ | $= r(-\sin\theta)$

$y_r = \frac{\partial}{\partial r}(r\sin\theta)$ | $y_\theta = \frac{\partial}{\partial \theta}[r\sin\theta]$

$y_r = \sin\theta$ | $= r\cos\theta$

$$\frac{\partial(x,y)}{\partial(r,\theta)} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix}$$

$(\cos\theta \times r\cos\theta) - (-r\sin\theta \times \sin\theta)$

$= r\cos^2\theta + r\sin^2\theta$

$= r(\cos^2\theta + \sin^2\theta)$

$= r(1)$

$= r$

Taylor series

Let $f(x,y)$ be a function defined in nbd of (a,b). Then, the taylor series expansion of $f(x,y)$ about the point (a,b) is taken as $f(x,y)$

$$f(x,y) = f(a,b) + \frac{1}{1!}\left[(x-a)f_x(a,b) + (y-b)f_y(a,b)\right] + \frac{1}{2!}\left[(x-a)^2 f_{xx}(a,b) + 2(x-a)(y-b)f_{xy}(a,b) + (y-b)^2 f_{yy}(a,b)\right] + \cdots \rightarrow (1)$$

Note:- * No need of 3rd degree terms.

* This series is called taylor series expansion of $f(x,y)$ in terms $(x-a)$ & $(y-b)$.

* If $(a,b) = (0,0)$ then from (1)

$$f(x,y) = f(0,0) + \frac{1}{1!}\left[x f_x(0,0) + y f_y(0,0)\right] + \frac{1}{2!}\left[x^2 f_{xx}(0,0) + 2xy f_{xy}(0,0) + y^2 f_{yy}(0,0)\right] + \cdots$$

this series is called as Maclaurin's series. (or) it is expansion of $f(x,y)$ in terms of x & y.

Session - 11

1) Express the taylor series expansion $f(x,y) = x^2y + 6y + 5x - 2$ in powers of $(x+1)$ and $(y-2)$ upto 2nd degree.

Sol:-

$$f(x,y) = f(-1,+2) + \frac{1}{1!}\left[(x+1)f_x(-1,2) + (y-2)f_y(-1,2)\right]$$

$$+ \frac{1}{2!}\left[(x+1)^2 f_{xx}(-1,2) + 2(x+1)(y-2)f_{xy}(-1,2) + (y-2)^2 f_{yy}(-1,2)\right] + \cdots$$

Given

$f = x^2y + 6y + 5x - 2$.

$(a,b) = (-1,2)$

$f(a,b) = (-1,2) = (-1)^2(2) + 6(2) + 5(-1) - 2$

$= \cancel{2} + 12 - 5 \cancel{-2}$

$= 7$

$f_x = \frac{\partial f}{\partial x} = \frac{\partial}{\partial x}(x^2y + 6y + 5x - 2)$

$= y(2x) + 5$

$\boxed{f_x = 2xy + 5}$

$f_y = \frac{\partial f}{\partial y} = \frac{\partial}{\partial y}(x^2y + 6y + 5x - 2)$

$= x^2(1) + 6$

$\boxed{f_y = x^2 + 6}$

$$f_{xx} = \frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial}{\partial x}\left(2x\,y+5\right)$$

$$= y.(2)$$

$$\boxed{f_{xx} = 2y}$$

$$f_{yy} = \frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial}{\partial y}\left(x^2+6\right)$$

$$f_{yy} = 0 + 0$$

$$\boxed{f_{yy} = 0}$$

$$f_{xy} = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x}\left[\frac{\partial f}{\partial y}\right]$$

$$= \frac{\partial}{\partial x}\left[x^2+6\right] = 2x$$

~~$f(x,y) = 7 + 1$~~

$$f_{x(a,b)} = f_x(-1,2) = 2xy+5$$

$$= 2(-1)(2)+5$$

$$= -4+5$$

$$\boxed{f(x)_{(a,b)} = +1}$$

$$f_{y(a,b)} = x^2+6$$

$$= (-1)^2+6$$

$$\boxed{f_{y(a,b)} = 7}$$

$$f_{xx(a,b)} = 2y$$

$$= 2(2)$$

$$\boxed{f_{xx(a,b)} = 4}$$

$$f_{yy(a,b)} = 0$$

$$f_{xy(a,b)} = 2x$$

$$= 2(-1) = -2$$

$$f(x,y) = 7 + \frac{1}{1!}\left[(x+1)(1) + (y-2)(7)\right] + \frac{1}{2!}\left[(x+1)^2(4) + 2(x+1)(y-2) + (y-2)^2(0)\right]$$

$$(-2)^2$$

$$= 7 + 1\left[\cancel{(x+1) + 7y - 14}\right] + \frac{1}{2}\left[(x^2+1^2+2x)4 + (2x+2)(-2y+4) + 0\right]$$

$$= 7 + \left[x + 1 + 7y - 13\right.$$

$$= 7 + \left[(x+1) + 7(y-2)\right] + \frac{1}{2}\left[4(x+1)^2 - 4(x+1)(y-2)\right] + \ldots$$

② Expand $f(x,y) = \sin x \cdot \cos y$ in powers of 'x' & 'y'.

$$f(x,y) = f(0,0) + \frac{1}{1!}\left[x f_x(0,0) + y f_y(0,0)\right] + \frac{1}{2!}\left[x^2 f_{xx}(0,0) + 2(x)(y) f_{xy}(0,0) + y^2 f_{yy}(0,0)\right].$$

$$f_x = \frac{\partial f}{\partial x} = \frac{\partial}{\partial x}\left[\sin x \cdot \cos y\right]$$

$$\boxed{f_x = \cos y \cos x}$$

$$f_y = \frac{\partial f}{\partial y} = \frac{\partial}{\partial y}\left[\sin x \cdot \cos y\right]$$

$$\boxed{f_y = -\sin x \cdot \sin y}$$

$$f_{xx} = \frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left[\frac{\partial f}{\partial x}\right] = \frac{\partial}{\partial x}\left[\cos x\right]$$

$$\boxed{f_{xx} = (-\sin x)\cos y}$$

$$f_{yy} = \frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y}\left[\frac{\partial f}{\partial y}\right] = \frac{\partial}{\partial y}\left[-\sin y \sin x\right]$$

$$f_{yy} = -\sin x \cos y$$

$$f_{xy} = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x}\left[\frac{\partial f}{\partial y}\right] = \frac{\partial}{\partial x}\left[-\sin y\right]$$

$$\boxed{f_{xy} = -\sin y \cdot \cos y}$$

$$f(x,y) = f = \sin x \cdot \cos y$$

$$f(0,0) = \sin(0) \cdot \cos(0)$$

$$\boxed{f(0,0) = 0}$$

$$f_{x(0,0)} = \cos y \cdot \cos x$$
$$\cos 0 \cdot \cos 0$$
$$\boxed{f_x = 1}$$

$$f_{y(0,0)} = -\sin y \cdot \sin x$$
$$f_y = -\sin(0) \cdot \sin 0$$
$$\boxed{f_y = 0}$$

$$f_{xx(0,0)} = -\sin x$$
$$= -\sin(0) \cdot \cos(0)$$
$$\boxed{f_{xx} = 0}$$

$$f_{xy} = -\sin y \cdot \cos y$$
$$= -\sin(0) \cdot \cos 0$$
$$\boxed{f_{xy} = 0}$$

$$f_{yy(0,0)} = -\sin x \cos y$$
$$= -\sin(0) \cdot \cos(0)$$

$$\boxed{f_{yy(0,0)} = 0}$$

$$f(x,y) = 0 + (x(1) + 0] + \frac{1}{2!}(0) + \cdots$$

$$= x + \cdots$$

//

Maxima & Minima

① Find p, q, r, s, t

$$\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial^2 f}{\partial x^2}, \frac{\partial^2 f}{\partial x \partial y}, \frac{\partial^2 f}{\partial y^2}$$

② Take $p = 0$ & $q = 0$ and solve them for getting stationary points (a,b) (c,d)...

③ At (a,b) find all p, q, r, s, t.

Case I :-

If $(rt - s^2) > 0$ then

(i) If $r_{(a,b)} > 0 \Rightarrow$ 'f' has minima value at $(a,b) = f(a,b)$.

(ii) If $r_{(a,b)} < 0 \Rightarrow$ 'f' has maxima value at (a,b) and $f(a,b)$ is max value.

Case II :-

* If $(rt - s^2)_{(a,b)} < 0$

then 'f' has neither maxima nor minima at (a,b).

→ Here (a,b) is called saddle pair.

Case III :-

* If $(rt - s^2)_{(a,b)} = 0$

the further investigation need for setting maxima/minima for 'f' at (a,b)

Session -12 & 13 13/09/25

(4) Determine Maxima & Minima for the function $f(x,y) = x^2+y^2+6x+12$

Sol:- $p = \frac{\partial f}{\partial x} = \frac{\partial}{\partial x}(x^2+y^2+6x+12)$

$$\boxed{p = 2x+6}$$

$q = \frac{\partial f}{\partial y} = \frac{\partial}{\partial y}(x^2+y^2+6x+12)$

$$\boxed{q = 2y}$$

$r = \frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right)$

$= \frac{\partial}{\partial x}(2x+6)$

$$\boxed{r = 2}$$

$s = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right)$

$= \frac{\partial}{\partial x}(2y)$

$$\boxed{s = 0}$$

$t = \frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right)$

$= \frac{\partial}{\partial y}(2y)$

$$\boxed{t = 2}$$

$p = 0$

$2x+6 = 0$

$x = -\frac{6}{2}$

$$\boxed{x = -3}$$

$q = 0$

$2y = 0$

$$\boxed{y = 0}$$

$(x,y) = (-3,0)$

$rt - s^2 = 2(2) - (0)^2$

$\boxed{rt - s^2 = 4}$

$\therefore rt - s^2 > 0$ then $r_{(-3,0)} > 0$ then the ~~min~~ given function has minimum value at $(-3,0)$ is '3'.

Q. Identify the maxima or minima for the function $f(x,y) = 2x^2 + 2xy + 2y^2 - 6x$.

Sol:- $p = \frac{\partial f}{\partial x} = \frac{\partial}{\partial x}(2x^2 + 2xy + 2y^2 - 6x)$

$\boxed{p = 4x + 2y - 6}$

$q = \frac{\partial f}{\partial y} = \frac{\partial}{\partial y}(2x^2 + 2xy + 2y^2 - 6x)$

$\boxed{q = 2x + 4y}$

$r = \frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right)$

$= \frac{\partial}{\partial x}(4x + 2y - 6)$

$\boxed{r = 4}$

$s = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right)$

$= \frac{\partial}{\partial x}(2x + 4y)$

$\boxed{s = 2}$

$t = \frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right)$

$= \frac{\partial}{\partial y}(2x + 4y)$

$\boxed{t = 4}$

$rt - s^2 = 4(4) - (2)^2$

$= 16 - 4$

$\boxed{rt - s^2 = 12}$

$p = 0$

$4x + 2y - 6 = 0$

$q = 0$

$2x + 4y = 0$

Solve p & q

$4x + 2y - 6 = 0$ — (1)

$2x + 4y = 0 \Rightarrow x + 2y = 0$

$x = -2y$

$4(-2y) + 2y - 6 = 0$

$-8y + 2y - 6 = 0$

$-6y = 6$

$\boxed{y = -1}$

$2x + 4(-1) = 0$

$2x = 4$

$\boxed{x = 2}$ $\boxed{(x, y) = (2, -1)}$

$\therefore rt - s^2 > 0$ so, $r_{(a,b)} = 4 > 0$ then the given function has minimum value at $(2, -1)$ is -6

Lagrange's method for maxima/minima for the given function

Consider the function 'f' with three variables $f(x,y,z)$ with some condition $\phi(x,y,z)=0$ to find maxima/minima for $f(x,y,z)$ we consider the following steps.

Step 1:- $F = f + \lambda\phi$ then find

$$\frac{\partial F}{\partial x}, \frac{\partial F}{\partial y}, \frac{\partial F}{\partial z}$$

Step 2:- Take $\frac{\partial F}{\partial x} = 0, \frac{\partial F}{\partial y} = 0, \frac{\partial F}{\partial z} = 0$

Step 3:- Solve $\frac{\partial F}{\partial x} = 0, \frac{\partial F}{\partial y} = 0, \frac{\partial F}{\partial z} = 0$ with the condition $\phi(x,y,z) = 0$ for obtaining the values of x, y & z.

$\therefore$ Then the maxima (or) minima for term 'f' is taken as $f(x,y,z)$.

3. Identify minimum value for the ~~fn~~ $x^2+y^2+z^2$, given that $xyz = 27$.

Sol:- $f = x^2+y^2+z^2$; $\phi = xyz - 27 = 0$

$$F = x^2+y^2+z^2 + \lambda(xyz-27)$$

$$\frac{\partial F}{\partial x} = \frac{\partial}{\partial x}\left[x^2+y^2+z^2+\lambda(xyz-27)\right]$$

$$\frac{\partial F}{\partial x} = 2x + 0 + 0 + \lambda(yz).$$

$$\boxed{\frac{\partial f}{\partial x} = 2x + \lambda yz}$$

$$\frac{\partial F}{\partial y} = \frac{\partial}{\partial y}\left(x^2+y^2+z^2+\lambda(xyz-27)\right)$$

$$\boxed{\frac{\partial f}{\partial y} = 2y + \lambda xz}$$

$$\frac{\partial F}{\partial z} = \frac{\partial}{\partial y}\left(x^2+y^2+z^2+\lambda(xyz-27)\right)$$

$$\boxed{\frac{\partial f}{\partial z} = 2z + \lambda xy}$$

$\rightarrow \frac{\partial f}{\partial x} = 0$

$2x + \lambda yz = 0$

$\lambda = \frac{-2x}{yz}$

$\frac{\partial f}{\partial y} = 0$

$2y + \lambda xz = 0$

$\lambda = \frac{-2y}{xz}$

$\frac{\partial F}{\partial z} = 0$

$2z + \lambda xy = 0$

$\lambda = \frac{-2z}{xy}$

$$-\frac{2x}{yz} = \frac{-2y}{zx} = \frac{-2z}{xy}$$

$$\frac{-\cancel{2}x}{y\cancel{z}} = \frac{-\cancel{2}y}{\cancel{z}x} \qquad \frac{-\cancel{2}y}{z\cancel{x}} = \frac{-\cancel{2}z}{\cancel{x}y}$$

$$\boxed{x^2 = y^2} \qquad \boxed{y^2 = z^2}$$

$$x^2 = y^2 = z^2$$

$$x = y = z$$

$x \cdot y \cdot z$

$x \cdot x \cdot x = 27$

$x^3 = 27$

$\boxed{x = 3}$; $\boxed{y = 3}$; $\boxed{z = 3}$

$\therefore$ min value is

$f(3,3,3) = 3^2 + 3^2 + 3^2$

$= 27$

7/09/25

Determine the dimensions of rectangular box of open end & top of max capacity with surface is 432 sq cm.

Sol:- Assume dimensions of open top as x, y and z.

~~Take~~ $V = xyz$
↓
volume of rectangle.

$2xy + 2yz + 2zx = S$ (Considering all the faces).

for if open top is removed then

$f = xyz$; $\phi = 2xy + 2yz + zx = 432$.
↓
open top is not considered.

Take $F = f + \lambda\phi$

$F = xyz + \lambda[2xy + 2yz + zx - 432]$

$$\frac{\partial F}{\partial x} = \frac{\partial}{\partial x}[xyz + \lambda(2xy + 2yz + zx - 432]$$

$$\frac{\partial F}{\partial x} = yz + \lambda[2y + z]$$

$$\frac{\partial F}{\partial y} = \frac{\partial}{\partial y}[xyz + \lambda(2xy + 2yz + zx - 432]]$$

$$\frac{\partial F}{\partial y} = xz + \lambda[2x + 2z)$$

$$\frac{\partial F}{\partial z} = \frac{\partial}{\partial z}[xyz + \lambda(2xy + 2yz + zx - 432)]$$

$$= xy + \lambda(2y + x)$$

$\rightarrow \frac{\partial F}{\partial x} = 0, \quad \frac{\partial F}{\partial y} = 0, \quad \frac{\partial F}{\partial z} = 0$

$yz + \lambda(2y + z) = 0 \Rightarrow \lambda = -yz/(2y+z)$

$xz + \lambda[2x + 2z] = 0 \Rightarrow \lambda = -xz/(2x+2z)$

$xy + \lambda(2y + x) = 0 \Rightarrow \lambda = -xy/(2y+x)$

$$\frac{-yz}{(2y+z)} = \frac{-xz}{2x+2z} = \frac{-xy}{2y+x}$$

$$\Rightarrow \frac{\cancel{-}y\cancel{z}}{2y+z} = \frac{\cancel{-}x\cancel{z}}{2x+2z}$$

$$\cancel{2xy} + 2zy = \cancel{2xy} + zx$$

$$\boxed{2y = x}$$

$$\Rightarrow \frac{\cancel{-}\cancel{x}z}{2x+2z} = \frac{\cancel{-}\cancel{x}y}{2y+x}$$

$$2\cancel{y}z + z\cancel{x} = 2\cancel{x}y + 2yz$$

$$\boxed{2y = z}$$

$$\boxed{\therefore x = 2y = z}$$

$y = \frac{x}{2}$ | $z = x$

$y = \frac{z}{2}$

$$\Rightarrow \cancel{2}x\left(\frac{x}{\cancel{2}}\right) + \cancel{2}\left(\frac{x}{\cancel{2}}\right)(x) + (x)(x) = 432$$

$$\cancel{3}x^2 = 43\cancel{2}$$

$$x^2 = 144$$

$$\boxed{x = 12}$$

Max capacity $= F(x, y, z)$

$= f(12, 6, 12)$

$= 12 \cdot 6 \cdot 12$

$= 864$

Solution of Higher order ODE :

* $\left(\frac{d^2y}{dx^2}\right) + k_1 \frac{dy}{dx} + k_2\, dy = \phi(x)$

this is the second order linear ODE with Constant coefficients 'k_1' and 'k_2'.

* If $\phi(x) = 0$, then the equation is said to be Homogeneous Linear O.DE

* If $\phi(x) \neq 0$ nonhomogeneous linear O.D.E

Take $D = \frac{d}{dx}$

$\Rightarrow D^2y + k_1 Dy + k_2 y = \phi(x)$

$[D^2 + k_1 D + k_2]y = \phi(x)$

$[F(D)]\, y = \phi(x)$

when

$F(D) = D^2 + k_1 D + k_2$

For solution of the eqn

The complete sol of ① is taken as

$\boxed{y = y_c + y_p}$

y_c : Complementary Function

Take Auxillary eqⁿ of the given eqⁿ is $f(m) = 0$ i.e; $m^2 + k_1 m + k_2 = 0$

$\Rightarrow$ By solving A.E, we get the roots. Let the two roots be m_1 & m_2.

Note

(1) If m_1 & m_2 are not equal and real. $\Rightarrow y_c = c_1 e^{m_1 x} + c_2 e^{m_2 x}$

(2) Both m_1 and m_2 are real $m_1 = m_2$ $\Rightarrow y_c = (c_1 + c_2 x) e^{m_1 x}$

(3) If $m_1 = \alpha + i\beta$ & $m_2 = \alpha - i\beta$ $\Rightarrow y_c = e^{\alpha x} [c_1 \cos\beta x + c_2 \sin\beta x]$

y_p : Particular Integral

$$y_p = \frac{1}{f(D)} \cdot \phi(x)$$

$\rightarrow$ If $F(x) = 0$, then $\boxed{y_p = 0}$

(1) $\phi(x) = e^{ax}$
$= e^{ax+b}$ $\Rightarrow y_p = \frac{1}{f(D)} e^{ax}$
$= \frac{1}{f(a)} e^{ax}$

(2) If $\phi(x) = \sin ax / \cos ax / \sin(ax+b) / \cos(ax+b)$

$\Rightarrow y_p = \frac{1}{f(D)} \sin ax = \frac{1}{f(D^2 = -a^2)} \sin ax$ if $f(D^2 = -a^2) \neq 0$

Ex:- $\frac{1}{D^2+3D+1}\sin 2x$

$$\boxed{= \frac{1}{-a^2+3D+1}\sin ax}$$

$= \frac{1}{-2^2+3D+1}\sin 2x$

$= \frac{1}{3D-3}\sin 2x$

$\frac{3D+3}{9D^2-9}\sin 2x$.

Session - 14

① Solve the diffential equation

$\frac{d^2y}{dx^2} - 9y = 0$.

Sol:- Take $D = \frac{d}{dx}$ ⇒ (operator form)

$= D^2 y - 9y = 0$

$[D^2 - 9]y = 0$

$[F(D)]y = 0$

$$\boxed{\therefore F(D) = D^2 - 9}$$

General solution of eqⁿ is

$$\boxed{Y = Y_c + Y_p}$$

*For $Y_c \Rightarrow F(m) = m^2 + k_1 m + k_2$

Here in this eq^n: $f(x)=0 \Rightarrow m^2-9=0$

$m^2=9 \Rightarrow m=\pm 3$

so $m_1=-3$, $m_2=3$

* m_1 & m_2 are real & different.

Now, $y_c = c_1 e^{-3x} + c_2 e^{3x}$

* As the eq^n is homogeneous $y_p=0$.

$\therefore y = y_c + y_p$

$\Rightarrow \boxed{y = c_1 e^{-3x} + c_2 e^{3x}}$

2) $y'' + y' - 2y = 0 \rightarrow (1)$; $y(0)=4 \rightarrow (2)$; $y'(0)=1 \rightarrow (3)$

Sol:- $\frac{d^2y}{dx^2} + \frac{dy}{dx} - 2y = 0$

$[D^2 + D - 2]y = 0$

$[F(D)]y = 0$ $\qquad \phi(x)=0.$

$\boxed{\therefore F(D) = D^2 + D - 2}$

$y_c \Rightarrow F(m)$ (Auxiallary eq^n).

m^2+m-2

$m^2+2m-m-2$

$m(m+2)-1(m+2)$

$(m-1)(m+2)$

$\boxed{m=1,-2}$

$m_1=-2$, $m_2=1$

$\therefore$ The roots are real and different.

$y_c = c_1 e^{-2x} + c_2 e^{x}$

$\rightarrow$ As $\phi(x) = 0$

$y_p = 0$

$\therefore y = y_c + y_p$

$$\boxed{y = c_1 e^{-2x} + c_2 e^{x}}$$

Given $y(0) = 4$ i.e;

$y = 4$ when $x = 0$

from (2), $y = c_1 e^{0} + c_2 e^{0}$

$y = c_1 + c_2$

$\boxed{c_1 + c_2 = 4} \rightarrow (4)$

from (3) $y'(0) = 1$

$y' = c_1 e^{-2x}(-2) + c_2 e^{x}$

$y' = -2c_1 e^{-2x} + c_2 e^{x}$

$y' = 1$ & $x = 0$

$1 = -2c_1 e^{0} + c_2 e^{0}$

$1 = -2c_1 + c_2$

$\boxed{-2c_1 + c_2 = 1} \rightarrow (5)$

By solving (4) & (5)

$c_1 + \cancel{c_2} = 4$

$\underline{\overset{+}{-}2c_1 \overset{}{+} \cancel{c_2} = \overset{}{-}1}$

$3c_1 = 3$

$\boxed{c_1 = 1}$

$c_1 + c_2 = 4$

$1 + c_2 = 4$

$\boxed{c_2 = 3}$

Now:- $\boxed{y = 1 \cdot e^{-2x} + 3e^{x}}$

③ Solve the differential equation 12/01/2025

$$\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6y = e^{2x}$$

Sol:- $D = \frac{d}{dx}$

$D^2y + 5Dy + 6y = e^{2x}$

$y[D^2+5D+6] = e^{2x}$

$[F(D)]\,y = e^{2x}$

$f(D) = D^2+5D+6$

The A.E is $f(m) = m^2+5m+6$

$\underline{m^2+3m} + \underline{2m+6}$

$m(m+3)+2(m+3)$

$(m+2)(m+3)$

$\boxed{m = -2, -3}$

real & different.

$m_1 = -3,\ m_2 = -2$

$\boxed{y = y_c + y_p}$

$\boxed{y_c = c_1e^{-3x} + c_2e^{-2x}}$

$\phi = e^{2x}$

exponential

$y_p = \frac{1}{f(D)}\phi(x) = \frac{1}{F(D)}e^{ax} \Rightarrow \frac{1}{f(a)}e^{ax}$

$= \frac{1}{D^2+5D+6}e^{2x}$

$f(2) = 2^2+10+6$

$f(2) = 20$

$= \frac{1}{20} e^{2x}$

General Solution is

$$\boxed{y = C_1 e^{-3x} + C_2 e^{-2x} + \frac{1}{20} e^{2x}}$$

(4) $\frac{d^3y}{dx^3} - \frac{3\,dy}{dx} + 2y = \sin x$.

Sol:- $D = \frac{d}{dx}$

$D^3 y - 3Dy + 2y = \sin x$.

$y\left[D^3 - 3D + 2\right] = \sin x$

$[F(D)]y = \sin x = \phi(x)$

$F(D): D^3 - 3D + 2 = 0 \Rightarrow$ A.E is $f(m) = m^3 - 3m + 2$

For $m = 1$ the eq^n becomes '0' so it is a root.

By synthetic divison

1	1	0	-3	2
	0	1	1	-2
	1	1	-2	0

$m^2 + m - 2 = 0$

$m^2 + 2m - m - 2 = 0$

$m(m+2) - 1(m+2) = 0$

$(m-1)(m+2)$.

$\boxed{m = 1, -2}$

$m_1 = 1, m_2 = 1, m_3 = -2$

$$\boxed{y = y_c + y_p}$$

$$\boxed{y_c = c_1 e^{x} + c_2 e^{x} + c_3 e^{-2x}}$$

$$\boxed{y_c = [c_1 + c_2 x] e^{x} + c_3 e^{-2x}}$$

$$y_p = \frac{1}{f(D)} \phi(x) = \frac{1}{D^2 - 3D + 2} (\sin ax)$$

$$= \frac{1}{f(D^2 = -a^2)} \sin ax \; \cancel{\frac{1}{-1^2 - 3D + 2}}$$

$$= \frac{1}{D^2 \cdot D - 3D + 2} \sin x$$

$$= \frac{1}{-(1)^2 \cdot D - 3D + 2}$$

$$= \frac{1}{-4D + 2} \sin x = \frac{1}{2[1 - 2D]} \sin x$$

$$= \frac{1}{2} \cdot \frac{1}{1 - 2D} \left(\frac{1 + 2D}{1 + 2D} \right) (\sin x)$$

$$= \frac{1}{2} \left[\frac{1 + 2D}{1 - 4D^2} \right] \sin x$$

$$= \frac{1}{2} [1 + 2D] \cdot \frac{1}{1 - 4D^2} (\sin x)$$

$$= \frac{1}{2} [1 + 2D] \cdot \frac{1}{1 - 4(-1)^2} (\sin x)$$

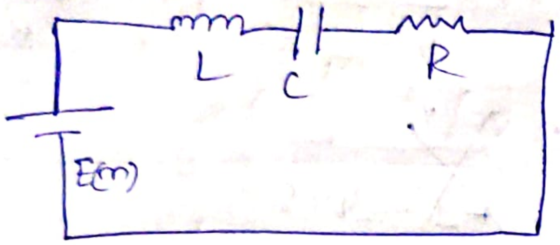
$$= \frac{1}{10} (1 + 2D)(\sin x).$$

$$= \frac{1}{10} [\sin x + 2D (\sin x)] = \frac{1}{10} \left[\sin x + 2 \frac{d}{dx} (\sin x) \right]$$

$$= \frac{1}{10}[\sin x + 2\cos x]$$

$$\boxed{y = [c_1 + c_2 x]e^{x} + c_3 e^{-2x} + \frac{1}{10}[\sin x + 2\cos x]}$$

LCR circuits



By Kirchoff's voltage law

$V_R + V_C + V_L = E(m) \rightarrow$ Electromotive force.

$$iR + \frac{q}{c} + L\frac{di}{dt} = Em$$

As $\Rightarrow i = \frac{dq}{dt}$

$$R \cdot \frac{dq}{dt} + \frac{q}{c} + L\frac{d}{dt}\left(\frac{dq}{dt}\right) = Em$$

$$\boxed{L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{q}{c} = Em}$$

$\Downarrow$

is the second order O.D.E corresponding to L.C.R circuit.

Note:-

* Differential eq^n corresponding to LC circuit is $\boxed{L\frac{d^2q}{dt^2} + \frac{q}{c} = E(m)}$

Session-15

1. Determine the charge correspond to LCR circuit with $L = 3H$, $R = 15\Omega$, $C = \frac{1}{12}F$, $E(t) = 0V$; $q(0) = 1C$; $i(0) = 2A$.

Sol:- $E(t) = 0$

w.k.t the differential equation corresponding to L.C.R circuit is

$$L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{1}{C}q = E(t)$$

$$3\frac{d^2q}{dt^2} + 15\frac{dq}{dt} + \frac{1}{1/12}q = 0$$

$$\frac{d^2q}{dt^2} + 5\frac{dq}{dt} + \frac{1}{\frac{1}{12}\times 3}q = 0$$

$$\frac{d^2q}{dt^2} + 5\frac{dq}{dt} + 4q = 0$$

$$D = \frac{d}{dt}$$

$$D^2q + 5Dq + 4q = 0$$

$$q[D^2 + 5D + 4] = 0$$

$$[f(D)]q = 0$$

$\Rightarrow$ A.E is $f(m) = m^2 + 5m + 4$

$$= \frac{-(5) \pm \sqrt{5^2 - 4(1)(4)}}{2(1)}$$

$$= \frac{-5 \pm \sqrt{25-16}}{2}$$

$$= \frac{-5 \pm \sqrt{9}}{2}$$

$$= \frac{-5 \pm 3}{2} = \frac{-5+3}{2} = \frac{-2}{2} = -1 \quad \Big| \quad \frac{-5-3}{2} = \frac{-8}{2} = -4$$

Both roots are real & different.

$q_0 = q_c + q_p$

$$\boxed{q_c = c_1 e^{-4t} + c_2 e^{-1t}}$$

$q_p = 0$ as $\phi(t) = 0$

$q = q_c + q_p$

$$\boxed{q = c_1 e^{-4t} + c_2 e^{-1t}}$$

(i) $q(0) = 1C$

$c_1 e^{-4(0)} + c_2 e^{-1(0)} = 1$

$$\boxed{c_1 + c_2 = 1} \rightarrow (1)$$

(ii) $i(0) = 2$

$i = \frac{dq}{dt} = \frac{d}{dt}\left(c_1 e^{-4t} + c_2 e^{-1t}\right) = 2$

$i = -4c_1 e^{-4t} - c_2 e^{-t} = 2$

$= -4c_1 e^{-4(0)} - c_2 e^{(0)} = 2$

$= \boxed{-4c_1 - c_2 = 2}$

$= \boxed{4c_1 + c_2 = -2} \rightarrow (2)$

Solve (1) & (2)

$c_1 + \cancel{c_2} = 1$

$4c_1 + \cancel{c_2} = -2$

$-3c_1 = 3$

$$\boxed{c_1 = -1}$$

$c_1 + c_2 = 1$

$-\frac{3}{5} + c_2 = 1$

$c_2 = 1 + \frac{3}{5}$

$\boxed{\cancel{c_2 = \frac{8}{5}}}$

$\boxed{c_2 = 2}$

$q = -\cancel{\frac{3}{5}e^{-4t}} - 1e^{-4t} + 2e^{-t}$

③ The motion of a mass spring system without damping is described by the initial value problem $\frac{d^2x}{dt^2} + 3\frac{dx}{dt} + 2x = e^{2t}$

$x(0) = 0$, $x'(0) = 2$

Sol:- $D = \frac{d}{dt}$

$[D^2 + 3D + 2]x = e^{2t}$

$[f(D)]x = \phi(t)$

where $f(D) = D^2 + 3D + 2$

$\phi(t) = e^{2t}$

$\boxed{x = x_c + x_p}$

$A.E \Rightarrow f(m) = 0 \Rightarrow m^2 + 3m + 2$

$m^2 + 2m + m + 2$

$m(m+2) + 1(m+2)$

$(m+1)(m+2)$

$\boxed{m = -1, -2}$

$\boxed{x_c = c_1 e^{-t} + c_2 e^{-2t}}$

$x_p = \frac{1}{f(D)} e^{2t}$

$= \frac{1}{D^2 + 3D + 2} e^{2t}$

$\boxed{x_p = \frac{1}{12} e^{2t}}$

$\boxed{x = c_1 e^{-t} + c_2 e^{-2t} + \frac{1}{12} e^{2t}}$

$x(0) = 0$

$$C_1 e^{-(0)} + C_2 e^{-2(0)} + \frac{1}{12} e^{2(0)} = 0$$

$$C_1 + C_2 + \frac{1}{12} = 0$$

$$\boxed{C_1 + C_2 = \frac{-1}{12}} \rightarrow 1$$

$x'(0) = 2$

$$-C_1 e^{-t} - 2 C_2 e^{-2t} + 2 \times \frac{1}{12} e^{2t} = 2$$

$$-C_1 e^{(0)} - 2 C_2 e^{-2(0)} + \frac{1}{6} e^{2(0)} = 2$$

$$-C_1 - 2C_2 + \frac{1}{6} = 2$$

$$-C_1 - 2C_2 = 2 - \frac{1}{6}$$

$$-C_1 - 2C_2 = \frac{12 - 1}{6}$$

$$\boxed{-C_1 - 2C_2 = \frac{11}{6}}$$

$$\boxed{C_1 + 2C_2 = -\frac{11}{6}} \rightarrow 2$$

Solve 1 & 2

$$\cancel{C_1} + C_2 = \frac{-1}{12}$$

$$\cancel{C_1} + 2C_2 = \frac{-11}{6}$$

$$- \quad - \quad +$$

$$-C_2 = \frac{11}{6} - \frac{1}{12}$$

$$-C_2 = \frac{22 - 1}{12} \Rightarrow -C_2 = \frac{21}{12}$$

$$\boxed{C_2 = -\frac{21}{12}}$$

$C_1 - \frac{21}{12} = \frac{-1}{12}$

$C_1 = -\frac{1}{12} + \frac{21}{12}$

$$\boxed{C_1 = \frac{20}{12}}$$

$$\boxed{x = \frac{20}{12} e^{-t} - \frac{21}{12} e^{-2t} + \frac{1}{12} e^{2t}.}$$

//.

Evaluate y_p for $\frac{d^2y}{dx^2} + 9y = \sin 3x$

Sol:-

$y_p = \frac{1}{f(D)} \sin 3x$

$D^2 y + 9y = \sin 3x$

$y \underbrace{[D^2 + 9]}_{f(D)}$

$\frac{1}{f(D)} = \frac{1}{D^2+9} \sin 3x$ Here $f(D^2 - a^2 = 0)$

So $\cancel{f(D^2 = -a^2)}$ we take $x \cdot \frac{1}{f'(D)} \sin ax$

$f'(D) = 2D$ if $F(D) \neq 0$.

$f''(D) = 2$

$\Rightarrow x \cdot \frac{1}{2D} \sin 3x$

$= \frac{x}{2} \cdot \frac{1}{D} (\sin 3x)$

$= \frac{x}{2} - \frac{\cos 3x}{3}$ $\boxed{= -\frac{x \cos 3x}{6}}$

//.